

THOMAS M

Durgapur, West Bengal, India

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Summary

Computer Science Engineer pursuing M.Tech in Operations Research at NIT Durgapur. Interested in backend development, algorithms, and optimization, with experience in Java, Spring Boot, Android development, and machine learning.

Education

Sri Sairam Institute of Technology

B.E. Computer Science and Engineering — CGPA: 8.53/10

2022 – 2026

Chennai, Tamil Nadu

National Institute of Technology Durgapur

M.Tech, Operations Research

2026 – 2028

Durgapur, West Bengal

Experience

Hexaind Technologies and Private Ltd

Jun 2024 – Sep 2024

AI/ML Intern (Remote)

- Spearheaded a real-time face and eye detection pipeline utilizing OpenCV vision frameworks.
- Processed and engineered a dataset of **500+ image samples** to fine-tune model reliability.
- Implemented targeted algorithmic optimizations to reduce frame processing latency by **15%**.
- Integrated individual modules into **1 cohesive system** engineered for automated vision tasks.

Projects

QuantMatch – Electronic Trading Platform | *Java, Spring Boot, MySQL, Redis, JWT, Spring Security*

- Engineered a high-performance financial exchange architecture utilizing **4 specialized backend modules**.
- Programmed a matching engine running a Limit Order Book based on strict Price-Time Priority rules.
- Automated the trade execution and tracking loop to log immutable updates across **3 database tables**.
- Integrated Redis Caching for fast snapshots, reducing repeated database lookup loads by **40%**.
- Exposed **7 secure REST API endpoints** protected under stateless JWT token validation filters.

TravelAssist – Roadside Assistance Platform | *Java, Android Studio, XML, Spring Boot, REST APIs, MongoDB*

- Architected a full-stack emergency response platform offering automated vehicle troubleshooting.
- Built an Android client delivering real-time service requests and **3 distinct user-tracking options**.
- Authored **10+ secure RESTful API endpoints** in Spring Boot to handle user authentication routing.
- Modeled and normalized **5+ MongoDB collections** to store multi-tier transaction records efficiently.

Technical Skills

Languages: Java, C++, C, Python

Backend: Spring Boot, REST APIs, JPA, Hibernate

Databases: MongoDB, MySQL, Firebase

Mobile Development: Android Studio, XML

Core Computer Science: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks

Tools: Git, GitHub, Postman

Achievements

- Qualified GATE Computer Science and Information Technology (CS) examination twice.
- Solved 440+ Data Structures and Algorithms problems on LeetCode.

Certifications

NPTEL: Cloud Computing, Internet of Things, Python for Data Science, Software Conceptual Design

MathWorks: Machine Learning Onramp, Deep Learning Onramp

MongoDB: Introduction to MongoDB, AI and Vector Search

Coursera: Business Analysis and Process Management